



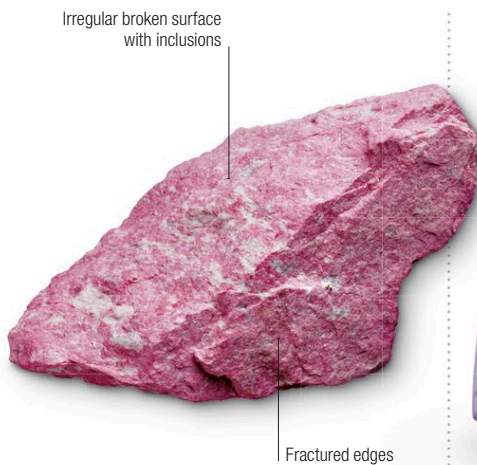
△ **Mixed-cut** tanzanite gemstone

Zoisite

The mineral name “zoisite” may not be particularly well known, but its gemstone variety, tanzanite, is popular among collectors. Found in lilac-blue to sapphire-blue colouring, it was named after its place of discovery, Tanzania. Tanzanite crystals have strong pleochroism, and show grey, purple, or blue depending on the angle from which they are viewed. Another, pink, variety is called thulite, from Thule, the name of an ancient island now thought to be Norway. Ordinary zoisite is usually massive and can be carved as a decorative stone, and as beads or cabochons.

Specification

Chemical name	Calcium aluminosilicate	Formula					
$\text{Ca}_2\text{Al}_3(\text{SiO}_4)_3(\text{OH})$	Colours	Blue, pink, white, light brown, green, grey	Structure	Orthorhombic	Hardness	6.5–7	
SG	3.2–3.4	RI	1.69–1.70	Lustre	Vitreous	Streak	White
Locations	Tanzania, Norway, Italy, Spain, Germany, Scotland, Japan						



Uncut thulite | Rough | Displaying a light pink colour and dense texture, this piece of thulite rough would be suitable for carving or cutting into cabochons.



Tanzanite rough | Rough | This piece of tanzanite rough shows excellent colour and transparency, and still retains much of its original crystal form.



Thulite cabochon | Cut | The colour of thulite tends to be subtle and soft. This thulite cabochon cut with a low dome has a delicate pink colour.



Huge tanzanite | Cut | Tanzanite gems over five carats are uncommon. This stunning, curved triangular brilliant-cut gem weighs 15.34 carats.



Cluster ring | Set | This ring features a cushion-cut tanzanite weighing 5.46 carats, encircled by a surround of brilliant-cut diamonds, with baguette diamonds on the openwork sides.

Brilliant-cut diamonds

Ruby in zoisite

Naturally occurring patterns

Anyolite is the name of a brilliant green variety of zoisite sprinkled through with rubies. The bright red rubies are often distorted and irregularly spread throughout the sea of massive green zoisite, and can vary in size from a few millimeters to several centimetres. The rubies are not of gem quality, but their colour provides a striking contrast to the green zoisite. It is popular as a carving and ornamental stone, with the rubies greatly enhancing the decorative pieces carved from it.



Red on green The contrast of the bright red ruby with the green zoisite is clear in this specimen.